

SMAP L-Band Microwave Radiometer: RFI Mitigation Prelaunch Analysis and First Year On-Orbit Observations

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Abstract—The National Aeronautics and Space Administration's (NASA) Soil Moisture Active and Passive (SMAP) mission, which was launched on January 31, 2015, is providing global measurements of soil moisture and freeze/thaw state. The SMAP radiometer operates within the protected Earth Exploration Satellite Service passive frequency allocation of 1400–1427 MHz. However, unauthorized in-band transmitters and out-of-band emissions from transmitters operating at frequencies adjacent to this allocated spectrum are known to cause interference to microwave radiometry in this band. Because measurement corruption by these terrestrial transmissions, which is referred to as radio-frequency interference (RFI), threatens mission success, the SMAP radiometer includes special flight hardware to enable the detection and filtering of RFI. Results from the first year of SMAP data show the presence of RFI with frequent occurrence over Asia and Europe. During the calibration/validation stage of the mission, the RFI detection and mitigation algorithms were modified to provide enhanced performance. Analysis of the L1B_TB products indicates good algorithmic performance with respect to RFI detection and removal. However, some regions of the globe (e.g., Japan) continue to experience complete data loss. This paper summarizes updates to the SMAP RFI processing algorithms based on prelaunch tests and on-orbit measurements, as well as RFI information obtained in SMAP's first year on orbit.

Index Terms—Microwave radiometry, radio-frequency interference (RFI).

I. INTRODUCTION

THE Soil Moisture Active and Passive (SMAP) radiometer, which was developed at the National Aeronautics and Space Administration's (NASA) Goddard Space Flight Center, was designed to measure ~1400- to 1424-MHz natural emission from the Earth's surface with ~40-km spatial resolution. The primary science objective of the radiometer is to provide soil moisture measurements with an uncertainty of $< 0.04 \text{ m}^3 \cdot \text{m}^{-3}$, which requires a radiometric uncertainty of $< 1.3 \text{ K}$ [1]. Anthropogenic sources, which are referred to as radio-

frequency interference (RFI), also impact the measurement. Calibration of SMAP's measurements is performed in a manner similar to other satellite radiometers [2], [3]; however, the SMAP ground processing algorithm includes RFI detection and mitigation stages [4] that are absent from other satellite radiometers.

Microwave radiometers are designed to be sensitive receivers and are thus susceptible to corruption from radio emissions from active services such as communication and navigation systems. Although the SMAP radiometer operates within the protected 1400- to 1427-MHz band, sources of RFI include illegal transmitters within the spectrum allocation and spurious or out-of-band emissions from transmitters operating in bands allocated for other active services. Prior to SMAP's January 2015 launch, results from the Soil Moisture and Ocean Salinity (SMOS) and Aquarius missions provided evidence of RFI globally, particularly over continental areas in Europe, Southeast Asia, and the Middle East [4], [5]. The SMOS team has continued alerting national spectrum management authorities concerning illegal transmissions resulting in the cancellation of many RFI sources [6]. Sources have been turned off in Europe, North America, and China; however, as expected and shown in SMAP data, the problem of RFI in microwave radiometer data still exists at L-band and will continue to be present as active services multiply and occupy the radio spectrum.

To combat the problem of RFI, particularly low-level RFI which can be concealed as natural variations in soil moisture, the SMAP radiometer includes an onboard digital detector back end with digital signal processing algorithms to produce measurements in time and frequency. The higher-than-conventional radiometer data volume allows for a combination of multiple detection algorithms to occur in ground processing. These algorithms were tested before launch using known RFI sources to measure performance. It will be shown that on-orbit performance agrees well with these prelaunch results.

II. RFI DETECTION ALGORITHMS

A description of the SMAP radiometer hardware and digital signal processing architecture is presented in [4]. The digital back end produces two independent data streams from radiometer observations referred to as “fullband” and “subband” data. For every pulse repetition interval (PRI) of the radar, the radiometer integrates for $\sim 300 \mu\text{s}$ during the receive window producing a fullband or a time-domain product covering the

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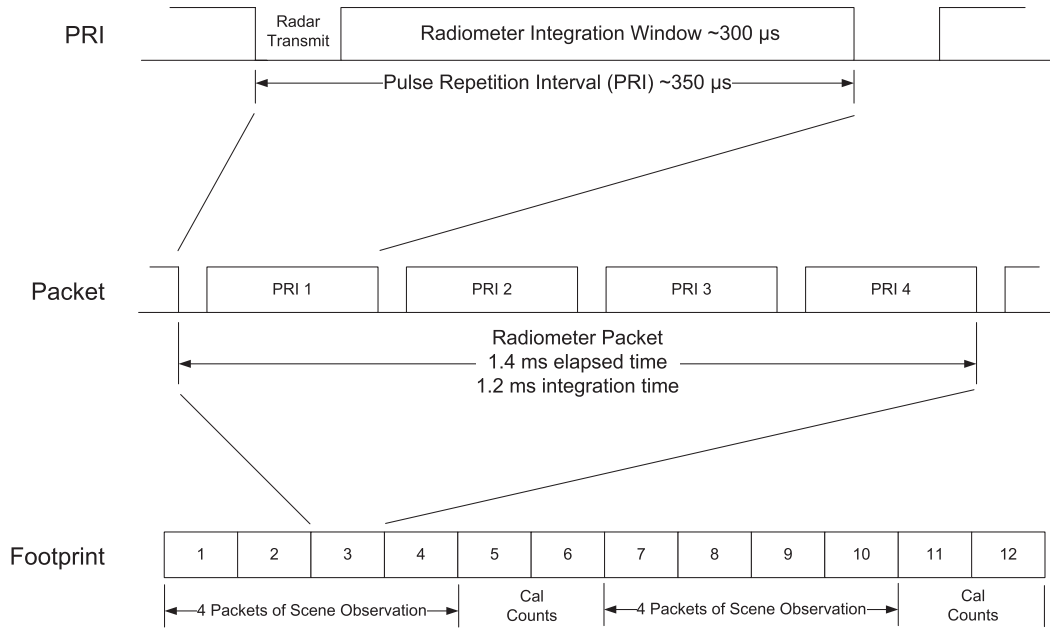


Fig. 1. Radiometer timing showing one SMAP radiometer footprint with eight packets for data measurement and four packets for calibration.

entire observed bandwidth. Radiometer packets are made up of four PRIs. Each science data packet includes fullband data for each of the four PRIs and subband data consisting of 16 frequency channels, which have been further integrated to four PRIs or ~ 1.2 ms (see Fig. 1). Note that 12 packets of data make up a footprint with eight packets of subband data or 32 PRIs of fullband data occurring for antenna observations. The remaining four packets are used to observe internal calibration references. The science data products downlinked for each fullband and subband channel include the first, second, third, and fourth integrated moments for both the horizontally and vertically polarized channels and separately expressed in terms of the in-phase and quadrature components of the signals and the complex cross-correlation coefficient used to compute the third and fourth Stokes parameters. The first and second moments are used to estimate the antenna brightness temperature of the vertical (T_V) and horizontal (T_H) channels, which are the primary science channels used by the soil moisture retrieval algorithm [7]. The third Stokes or the T_3 channel measurement is used to obtain the correction of Faraday rotation caused by the ionosphere [8], whereas the fourth Stokes or the T_4 channel is used to detect RFI.

The RFI detection algorithms in the L1B_TB ground processing include “pulse,” “cross-frequency,” “kurtosis,” and “polarimetric” approaches. The pulse and cross-frequency detectors operate on the fullband and subband data, respectively, whereas the kurtosis and polarimetric detectors operate on both. The pulse and cross-frequency detectors search for significant departures of brightness temperatures from the mean value of the scene under observation. These detection algorithms operate by flagging pixels which exceed a threshold, where the threshold is set at β standard deviations away from the mean. Instead of computing the standard deviation of the brightness temperature from the data themselves, as was done in the baseline algorithms reported in [4], an equation which

uses the estimated system temperature is used to determine the standard deviation as a function of the estimated mean. This procedure reduces variability in setting the thresholds for these detectors. Although RFI will produce a positive bias in the brightness temperatures, these detectors were made double sided to minimize any bias due to false alarms. These changes were made before launch and are used in the processing of on-orbit data.

The baseline pulse detection algorithm empirically calculated the standard deviation $\sigma_{td}(t)$ from the lowest 90% of samples within a specified time window. The finite number of samples used in this estimation resulted in a noisy estimate of the signal standard deviation and, therefore, caused a widely varying detection threshold from footprint to footprint. Widely varying thresholds can cause excessive false alarms and/or missed detections. The modified approach uses the following equation based on the model for computing the radiometer noise equivalent delta temperature (NEDT):

$$\sigma_{td}(t) = \frac{T_{rec} + m_{td}(t)}{\sqrt{BW * \tau}} \quad (1)$$

where T_{rec} is the radiometer receiver temperature, $m_{td}(t)$ is the robust estimate of the antenna temperature of the scene, BW is the radiometer bandwidth of 24 MHz, and τ is the integration time of the time sample. The mean of measured samples in the time window is estimated excluding the largest and smallest 5% of samples. Footprint samples that are β_{td} standard deviations above or below the mean are flagged.

The cross-frequency algorithm operates on a single time subsample of 16 frequency subbands at a time. The equation used to obtain the standard deviation for that time period is as follows:

$$\sigma_{cf}(t) = \frac{T_{rec} + m_{cf}(t)}{\sqrt{\frac{BW}{16} 4n\tau}} \quad (2)$$

where n is the number of time subsamples to be integrated within a footprint before cross-frequency detection ($n = 1$ or 8). For a given integration period, the mean $m_{cf}(t)$ is the robust estimate of the antenna temperature. It is estimated excluding the largest and smallest two channels. T_{rec} , BW, and τ are the same as previous. Any channel which contains an antenna temperature value that is β_{cf} standard deviations above or below the mean is flagged.

The kurtosis, which is calculated using the first four moments of the science telemetry, is used to distinguish Gaussian-distributed signals (such as thermal noise, for which the kurtosis is equal to 3) from non-Gaussian-distributed signals frequently characteristic of RFI [9]. Detection using kurtosis occurs if the absolute difference between the kurtosis observation and the nominal kurtosis value is greater than β_K standard deviations. The nominal kurtosis is determined *a priori* using RFI-free data. This detection algorithm is double sided, since kurtosis can go above or below 3 depending on the RFI type. The polarimetric detectors use a simple thresholding algorithm which searches for variations greater than a fixed number of standard deviations. These are also double sided, since RFI can cause T_3 and T_4 to be positive or negative. The cross-frequency and subband kurtosis algorithms blank frequency channels adjacent to those flagged, since RFI can spill over to adjacent channels.

The outputs of these detectors are combined in a maximum probability of detection (MPD) algorithm to remove corrupted pixels in each footprint. Each footprint consists of an 8-time-by-16-channel antenna temperature spectrogram obtained from subband data. The MPD array for each footprint is obtained by performing a logical OR of all the RFI detection flags from each detection algorithm. To incorporate the time-domain RFI detection flags into the MPD array, each time-domain flag is used to flag the corresponding time subsample in the frequency domain, essentially flagging all subbands in the MPD array [4].

Each detection algorithm has a detection threshold setting that determines its sensitivity and false alarm rate. Detection thresholds for each detector are specified in an ancillary settings file with a global spatial grid of adaptable resolution, distinct for ascending/descending passes and fore/aft looks. All thresholds are currently at a 0.1° spatial resolution. All detector thresholds are uniform in space and for the ascending/descending/fore/aft cases, with the exception of the pulse and T_4 detectors. Upon analysis of initial observations, the pulse detector thresholds were increased for coastal regions, since this detector may mistakenly detect coastal crossings as RFI due to the rapid variations between land and water crossings (see Fig. 2 for the time-domain spatial threshold settings at a 0.1° resolution). These settings remain constant over time and antenna position. The T_4 thresholds for both the fullband and subband data are similarly set, since the fourth Stokes are increased at the coastlines due to mismatches in the horizontally and vertically polarized antenna patterns. Note that the kurtosis does not suffer from this problem, since kurtosis values are insensitive to antenna temperature changes. The cross-frequency algorithm operates on a single time subsample at a time; thus, rapid antenna temperature variations with time, associated with land water crossings, do not affect its performance. Both the fullband and subband third Stokes detector settings are currently set to very

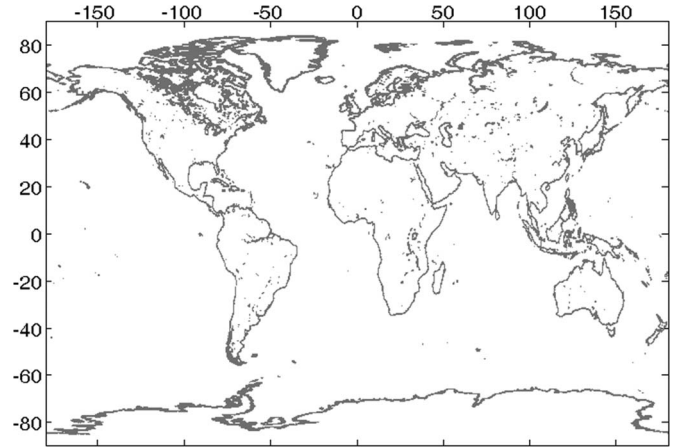


Fig. 2. Time-domain thresholds at 0.1° spatial resolution. The thresholds are increased to 5 standard deviations (β) away from the mean over coastal regions (gray) and 3 standard deviations from the mean otherwise (white).

high values, so that the third Stokes detectors are “turned off” and contribute only minimally to the overall false alarm rate.

III. PRELAUNCH RFI TESTING

To test the performance of SMAP RFI detection algorithms, a comprehensive test was conducted at the NASA Goddard Space Flight Center between April 6, 2013 and April 20, 2013 with the SMAP radiometer in thermal vacuum conditions. The objectives were to verify that SMAP can detect and mitigate pulsed and continuous-wave (CW) RFI and to determine the performance of RFI algorithms against low- and high-level RFI. During the test, known RFI signals were injected to the radiometer front end for a certain period of time (~ 30 s), followed by RFI-free states lasting for the same amount of time. Therefore, RFI detection performance could be evaluated by observing the difference between SMAP products after RFI mitigation for RFI-injected states and SMAP products for RFI-free states with and without mitigation, as described in [4]. CW and pulsed RFI signals with different frequencies and amplitudes were injected into the radiometer. The prelaunch results show SMAP performance against these RFI signals.

A. Performance of RFI Detectors

The performance of the RFI detection algorithms was evaluated using the receiver operating characteristic (ROC) curve and the associated area under the curve (AUC) calculated for each ROC curve. ROC curves indicate the performance of a detector by showing the probability of detection as a function of false alarm rate [10]. By varying a detector’s threshold over a large range, the associated probability of detection p_d and probability of false alarm p_{fa} can be computed and plotted. Since infinite and zero thresholds mean no detection and 100% detection, respectively, all ROC curves start from $(p_{fa}, p_d) = (0, 0)$ and end at $(p_{fa}, p_d) = (1, 1)$. The AUC is calculated by taking the integral of the ROC curve over the false alarm rate [11]. A “random guessing” detector has equal probabilities of false

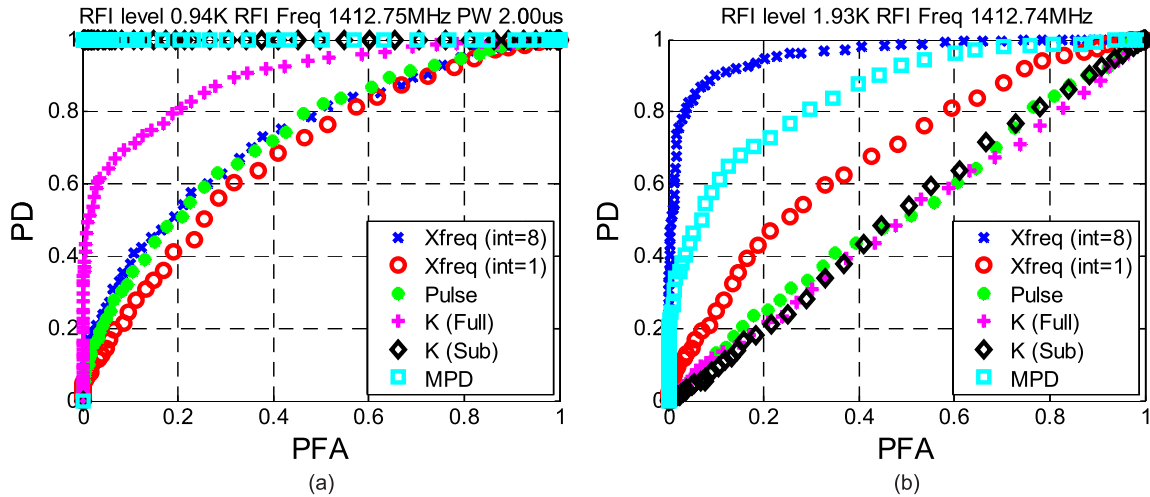


Fig. 3. Performance results for the pulse and cross-frequency detection (for both one- and eight-packet integration) and kurtosis fullband and subband detection algorithms for (a) pulsed RFI with a pulsewidth of 2 μ s and (b) CW RFI.

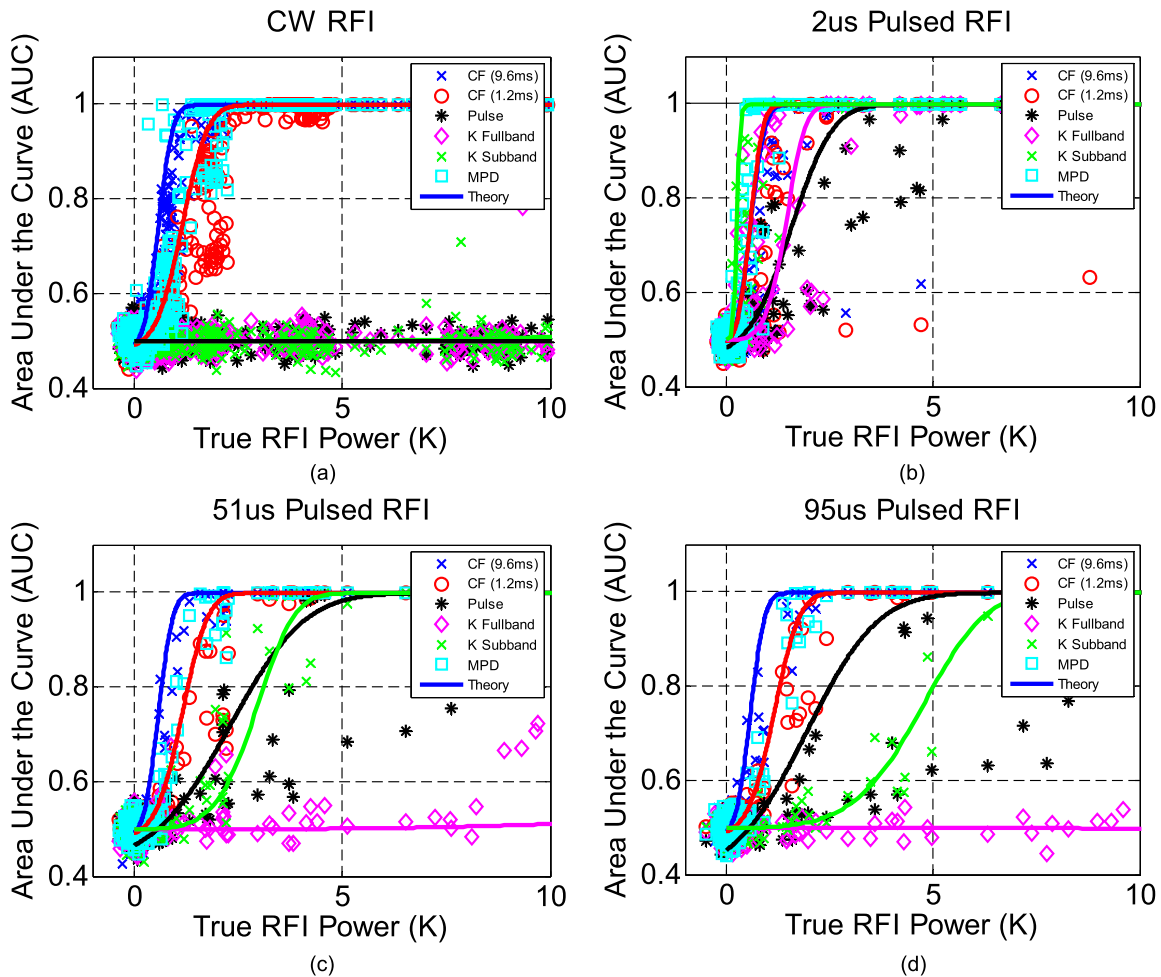


Fig. 4. AUC performance results for the cross frequency of 9.6-ms integration time, cross frequency of 1.2-ms integration time, pulse detection, kurtosis fullband, and kurtosis subband detection algorithms against a sweep of (a) CW and (b)–(d) pulsed sources.

alarm and detection, in which case the AUC is 0.5. Since a good detector maximizes the probability of detection even for small false alarm rates, higher AUC values mean higher probabilities

of detection over the same false alarm rates and thus better detectors. The ideal case is to obtain $p_d = 1$ for all false alarm rates, so that AUC = 1.

B. ROC Curves for Pulsed and CW RFI

To plot ROC curves and compute the AUC parameters against each RFI signal, RFI detection algorithms were applied to the measured data by changing the detector thresholds β in postprocessing. RFI detection is defined as any pixel flagged in a footprint. The footprint can be either the 8×16 spectrogram of subband data or 32 samples of fullband data. False alarm rates were computed by averaging the detection rates over 30 s of RFI-free measurement periods, and the corresponding probability of detection rates was calculated over the adjacent 30 s of RFI-injected measurement periods. RFI level or true RFI power is the difference between the average antenna temperature value of data with RFI and the average antenna temperature value of data without RFI.

ROC curves were generated for five different RFI baseline detectors: the cross frequency for integration times 1.2 ms (one-packet integration) and 9.6 ms (eight-packet integration), pulse detection, kurtosis fullband, and kurtosis subband. The polarization detectors were excluded from this analysis due to the symmetrical injection during the test of RFI into both the vertical and horizontal ports (an unrealistic RFI scenario). Fig. 3(a) illustrates the ROC curves for the detectors for pulsed-wave injected RFI of 2- μ s pulsewidth at 1412.75 MHz and RFI level of about 1 K. Pulses of 2- μ s width were injected at 365-Hz pulse repetition frequency; thus, each footprint has approximately four pulses, and each pulse falls in a different time sample. These narrow pulses localized in short-time and narrow-frequency intervals cause high-amplitude power spikes even for low-power RFI cases and are detected by almost all detectors even at power levels less than 1 K. The subband kurtosis algorithm is the most sensitive to such low-duty-cycle pulsed RFI, and all algorithms achieve good performance as the RFI power level increases. Fig. 3(b) illustrates measured ROC curves for CW RFI signals at 1412.75 MHz. The cross-frequency algorithms are more sensitive to single-tone CW interference than the kurtosis and pulse detectors [12]. Cross-frequency detection at 9.6-ms time resolution is observed to be the most effective algorithm in this case, since averaging over time reduces the noise fluctuations before detection.

Results from the combined MPD approach are also included. The MPD performance can differ from that of the best individual detector because it experiences a higher false alarm rate when it combines multiple detector outputs. This is a direct result of the MPD algorithm where flags from individual algorithms are combined with a logical OR operator.

C. AUC Analysis

The previous ROC curve examples showed RFI detection performance for specific RFI sources. In addition, included in the RFI test are pulsewidths of 51 and 95 μ s, which were injected at 748-Hz and 1-kHz pulse repetition rates, respectively. The AUC can be computed for each of these ROC curves, and results are then plotted as a function of the RFI source amplitude. Fig. 4 demonstrates such measured AUC values versus true RFI amplitudes for the CW and 2- μ s, 51- μ s, and 95- μ s pulsed RFI cases. The cross-frequency algorithms were found effective in all RFI cases examined (due to the modulated

sinusoidal sources considered), whereas pulse detection and the kurtosis algorithms lost their efficiency against continuous and high-duty-cycle pulsed RFI cases. The results in general confirm the ability of the SMAP RFI detection and mitigation algorithms to detect RFI at levels of 1–2 K.

It is possible to compute AUC values against these RFI signals theoretically under certain assumptions, as described in [13] and [14]. Predictions of the theory are shown as solid lines in Fig. 4. To compute AUC values theoretically, it was assumed that the 300- μ s integration intervals for fullband and the 1.2-ms integration intervals for subband detectors could not contain more than one RFI pulse and that RFI pulses existed completely within one integration period. It is also implicit that, in each 1.2-ms subband integration period, any present RFI pulse exists exactly at the center frequency of one of the 16 frequency channels and that the integration period is sufficient for the large sample limit of the kurtosis distribution to be applied [15].

Under these assumptions, 300- μ s fullband and 1.2-ms subband RFI-free measurements were regarded as central chi-squared distributions, whereas RFI-contaminated ones were considered to be noncentral chi-squared distributions, with the noncentrality parameter determined by the properties of the RFI signal, as explained in [13] and [14]. The kurtosis mean and standard deviation for RFI-free and sinusoidal RFI-contaminated measurements were determined as in [15]. Theoretical results for the AUC for the corresponding sources are included in Fig. 4 along with the measured data. It is observed that theoretical AUC values are generally slightly higher than the empirical results. This difference can be explained by the violation of the assumptions during the RFI test. RFI injection was not synchronized with SMAP radiometer timing; thus, RFI pulses are frequently split into more than one integration period. In addition, RFI frequencies do not align perfectly with SMAP subband channel frequencies. These two facts degrade the measured performance because RFI power is less detectable when split into multiple time/frequency samples instead of concentrated in a single time sample. Moreover, kurtosis detectors also show significant differences from the model, since the number of measurement samples used to compute kurtosis is not high enough to completely satisfy the central limit theorem conditions assumed in [15]. The 300- μ s and 1.2-ms integration intervals include 28 800 and 7200 samples for Nyquist sampling of a 24-MHz bandwidth, respectively. In general, the theoretical and empirical results match each other well given the inherent assumptions in the comparison.

IV. SMAP RFI OBSERVATIONS

Following its January 2015 launch, SMAP data products were analyzed to observe brightness temperatures and their RFI properties. RFI was observed in antenna brightness temperatures (TA) and is shown in Fig. 5. The figure is a peak hold plot at 0.25° resolution of horizontally polarized footprints taken from the period of June 3 to 9, 2015. The color scale is in kelvins and was limited to 300 K to better illustrate the RFI events, which appear as red (over land) in the figure. Strong RFI events occur globally but with increased probability spatially and temporally over Europe, the Middle East, and

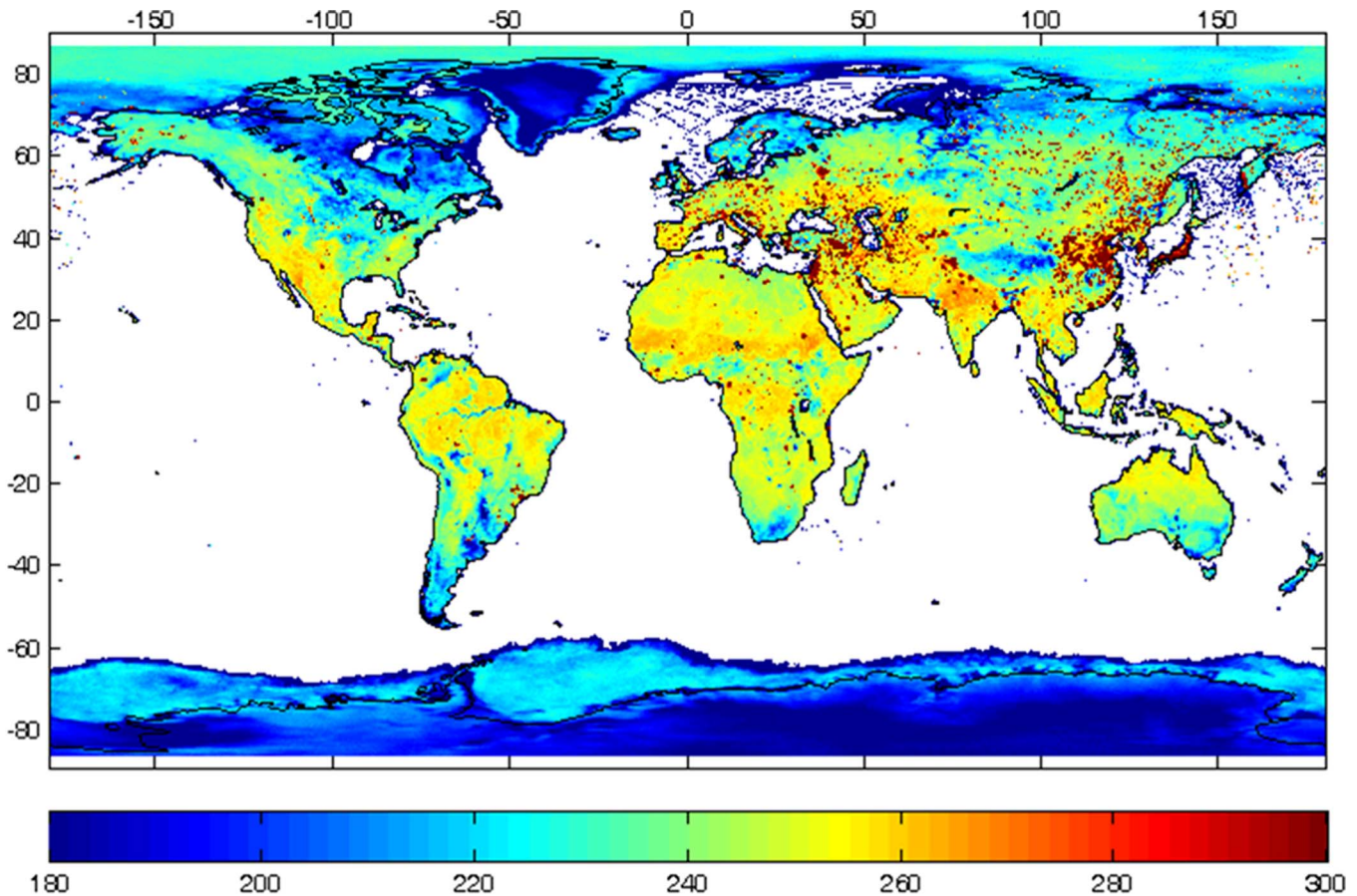


Fig. 5. Peak hold data on a $0.25^\circ \times 0.25^\circ$ global grid of horizontally polarized antenna temperatures (TA) in kelvins before RFI filtering for the period June 3, 2015 to June 9, 2015.

Asia. Fig. 6 illustrates the same data set after RFI filtering using the detection algorithms described in previous sections. The results demonstrate the effectiveness of the RFI detection and mitigation algorithms. RFI corruption is significantly reduced; however, there are still residual RFI which remain in the data over certain regions. An extensive area of Japan remains unrecoverable due to strong RFI events corrupting the entire footprints of the data. Missing data show up as white in the images. Ocean data ($TA < 100$ K) are omitted in Figs. 5 and 6 to better illustrate the RFI events which occur mostly over land.

The SMAP processor reports the antenna temperature both before and after RFI mitigation is applied. The difference between these quantities represents the detected RFI level. In cases where 100% of the spectrogram is flagged as RFI, the RFI mitigated antenna temperature is reported as a fill value (no valid output), and the RFI level cannot be computed in these instances. Fig. 7 shows plots of images of horizontally polarized RFI using data from the same June time frame. The two images were created using data from ascending and descending passes for forward looks. Differences between the images illustrate variations in RFI source transmissions with time of day and azimuth angle. The color scale was limited to 10 K to better observe RFI less than this value. RFI levels are slightly increased over coastlines of sea ice in the Arctic, which are a direct result of erroneous flagging by the time-domain detector (see Fig. 2 for the time-domain threshold

grid which does not include increased thresholds over sea ice boundaries). Only fullband data exist over the ocean for the data set shown in Fig. 7. As a result, the MPD algorithm previously described utilizes only the fullband detectors (pulse, kurtosis, T_4), resulting in a lower false alarm rate and thus a lower RFI level over ocean compared with land. Detector thresholds are uniform globally (except coastlines for the pulse and T_4 detectors); thus, RFI levels due to false alarm rate are also weakly proportional to the scene brightness temperature.

The radiometer L1B_TB product contains a quality flag indicating the quality of the brightness temperature measurement based on various corrections in the algorithm [16]. The quality flag includes subflags for an excessive degradation in radiometer sensitivity and an excessive detected RFI level. If the NEDT for the brightness temperature measurement exceeds 2 K, then this subflag is set. If the detected RFI level exceeds 100 K, this subflag is set, since there could be residual RFI in the footprint due to hardware limitations of interchannel isolation between every two channels. All subflags contribute to the master bit setting which, in turn, can be used to further filter data. The RFI subflags were therefore used to exclude data deemed to be of insufficient quality in the images in Fig. 8. Following this exclusion, regions such as Japan show almost no remaining measurements. Fig. 8(a) shows a zoomed-in image of Eastern Asia in Fig. 5. The results show strong RFI in Eastern China and over almost the entirety of Japan. There

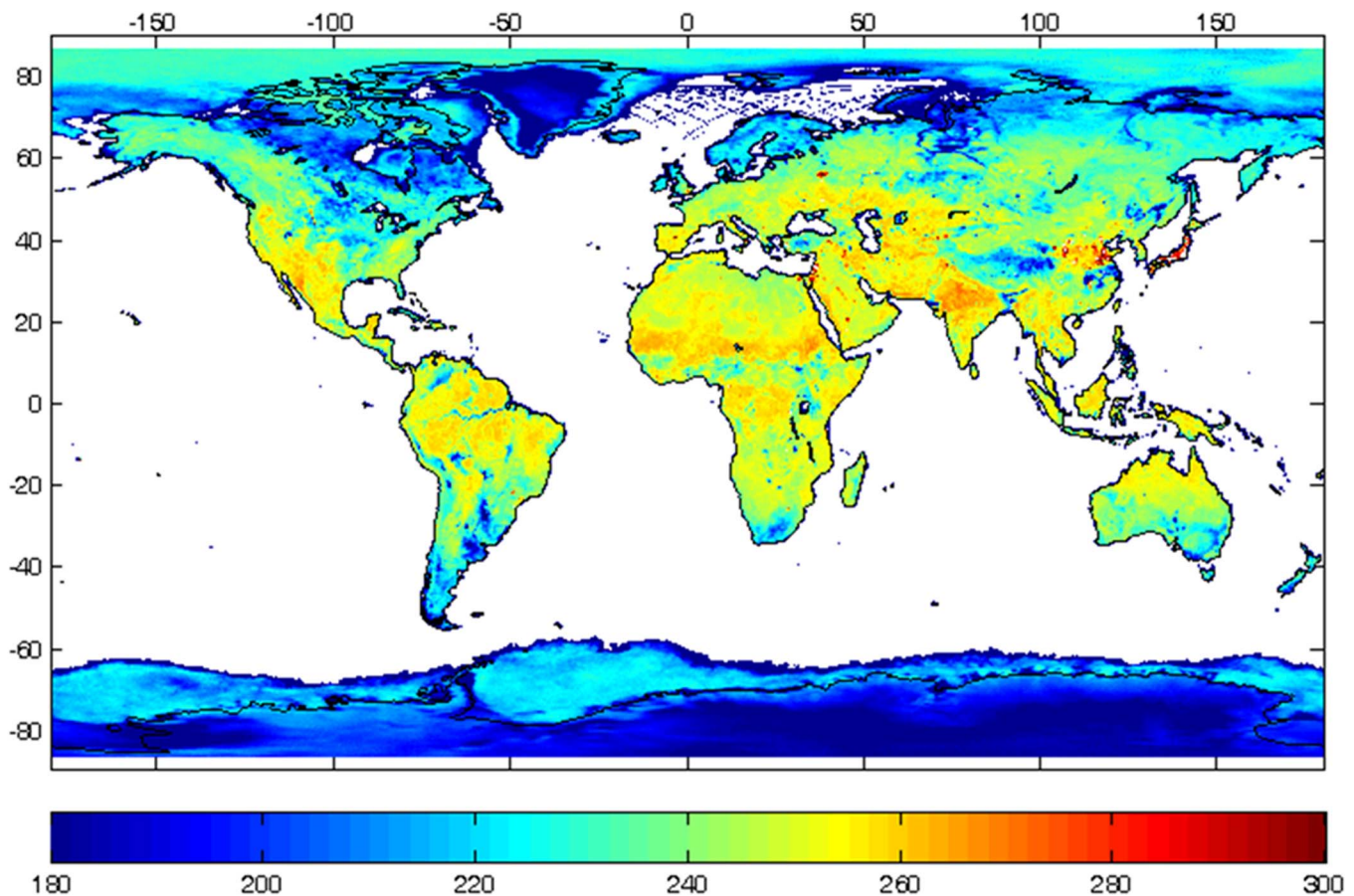


Fig. 6. Peak hold data on a $0.25 \times 0.25^\circ$ global grid of horizontally polarized antenna temperatures (TA) in kelvins after RFI filtering for the period June 3, 2015 to June 9, 2015.

is substantial reduction in corruption following RFI filtering shown in Fig. 8(b), but Japan remains significantly impacted by RFI. Fig. 8(c) illustrates the peak hold of RFI filtered data and excluding any points removed by data quality flags. These flags remove much of the data over Japan, showing that data containing possible residual RFI are at least being excluded from further analysis in most cases. Initial results indicate that these sources are wideband and occupy much of SMAP's bandwidth at all times (i.e., the entire 128-pixel spectrogram is corrupted.) Such wideband cases cannot be corrected by any of SMAP's RFI filtering procedures.

For a more detailed analysis, Fig. 9 contains data from a number of footprints of horizontally polarized antenna temperatures over Honshu, Japan. The plots in Fig. 9(a) and (b) illustrate geolocated footprints before and after RFI mitigation, respectively. Fig. 9(c) shows a frequency–time plot of the data before RFI filtering and footprint averaging. Fig. 9(d) shows the unaveraged data after the MPD algorithm detects and removes RFI. Pixels removed are white. The color scale was limited to 400 K in the plots with RFI to show the varying antenna temperatures over the scan. It is evident that the RFI covers a significant portion of the SMAP bandwidth. The cross-frequency algorithms are able to detect most of this RFI; in most time samples, all subbands are flagged due to adjacent channel blanking. The pulse and kurtosis detectors are insensitive to the wideband cases shown in the magenta ellipses. The

strong pervasive RFI makes it difficult to recover data, since the algorithm removes all pixels in numerous footprints. Any footprint contaminated by these wideband sources that has been recovered likely contains residual RFI. The pulse and cross-frequency detectors compute detection thresholds using an estimated mean of the scene; thus, sources that are persistent in time and/or frequency within a footprint bias their robust mean estimates high, leading to a reduction in detection sensitivity. The RFI that appears over the ocean (turquoise ellipses) is possibly from land-based emitters received through the antenna sidelobes. The pulse and kurtosis detectors are very sensitive to these and detect them easily as do the cross-frequency detectors. Due to the conservative approach of the MPD algorithm, these pulses detected in the time domain result in the removal of all data within the subbands of the corresponding time samples.

A. Kurtosis

In addition to RFI detection, the measured kurtosis information can be used to classify RFI types, at least in part. Kurtosis values below 3 typically indicate CW signals or pulsed RFI with a duty cycle greater than 50% relative to the integration time of the radiometer time sample. Pulsed RFI with duty cycle less than 50% produces kurtosis values greater than 3. Fig. 10 shows the fullband excess kurtosis map of RFI with short (red) and

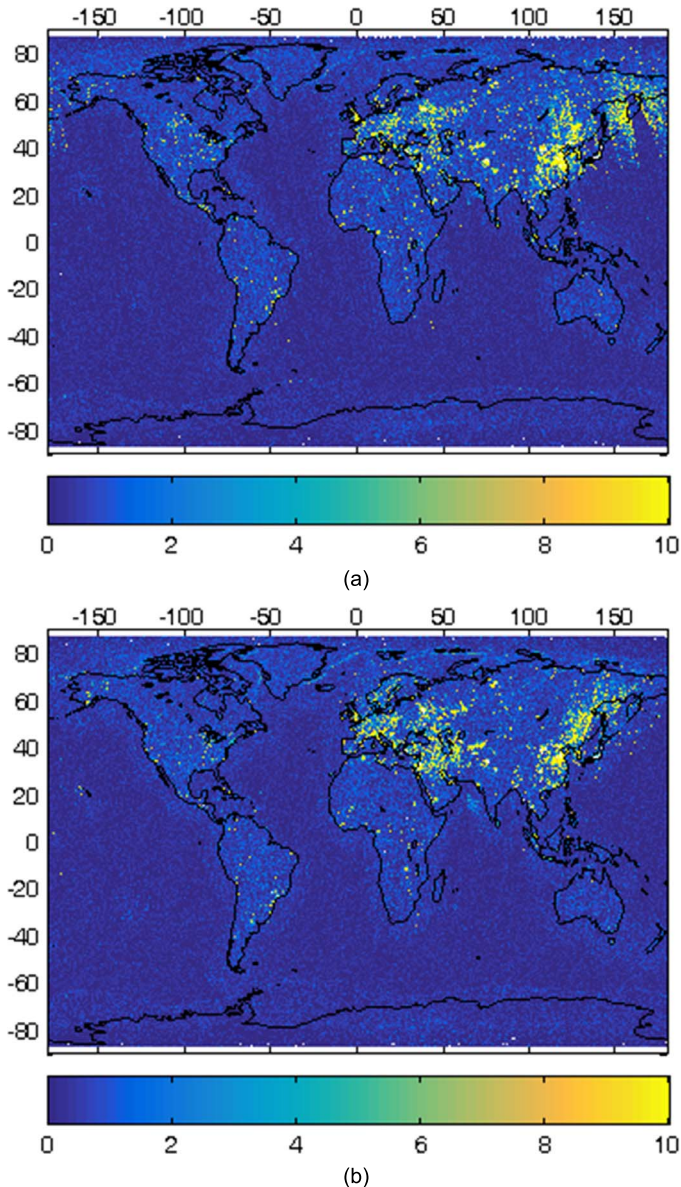


Fig. 7. Peak hold data on a $0.25^\circ \times 0.25^\circ$ global grid of horizontally polarized RFI in kelvins for the period June 3, 2015 to June 9, 2015 for (a) ascending and (b) descending passes, forward looks.

long (blue) duty cycles. The figure contains values of maximum and minimum kurtosis values minus 3. The map was created by overlaying peak kurtosis and minimum kurtosis values at 0.25° resolution. Global RFI shows both types, with pulsed RFI apparently being more common. The radars of the distant early warning line in the far north Arctic region of Canada show up prominently. Kurtosis is quite sensitive to these short duty cycle pulses. The CW RFI is distributed mostly over Asia but also appears in the Middle East and Europe. It is important in interpreting these results to recall that the kurtosis sensitivity to CW RFI is significantly reduced, as compared with short pulsed RFI of the same amplitude, so that a smaller apparent coverage of CW sources is not surprising. Over much of Japan, the fullband kurtosis does not appear to deviate significantly from the nominal value, a result due to the widespread wide-

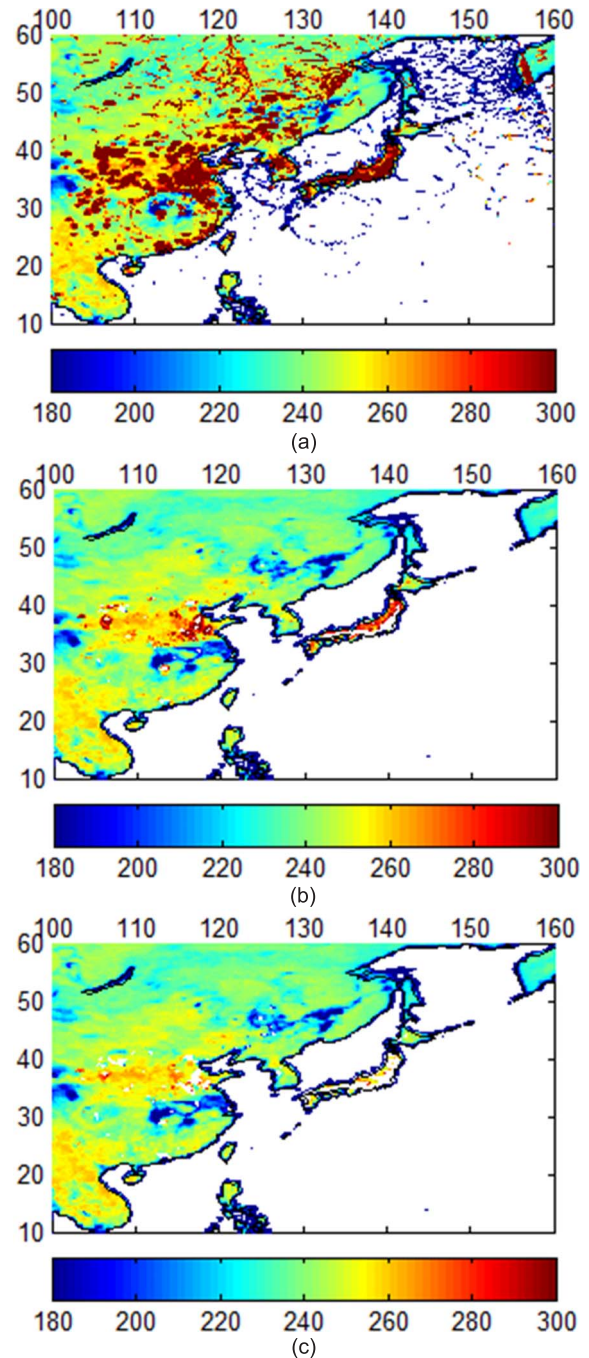


Fig. 8. Zoomed-in image of Eastern Asia in Fig. 5. Images are peak holds of horizontally polarized antenna temperatures (TA) in kelvins (a) prior to RFI filtering, (b) after RFI filtering, and (c) after RFI filtering and excluding any flagged data for the period June 3, 2015 to June 9, 2015. Ocean data (TA < 100 K) are omitted.

band cases over this region and the density of interferers within a footprint [17], making the data appear more Gaussian in nature. Although the subband kurtosis is more sensitive than the fullband kurtosis, it exhibits a similar behavior over this region. Fig. 11 shows plots of excess kurtosis values and detection rates at a 0.25° resolution. The subband excess kurtosis plot shown in Fig. 11(c) contains the maximum and minimum values among all 16 subbands in a given footprint.

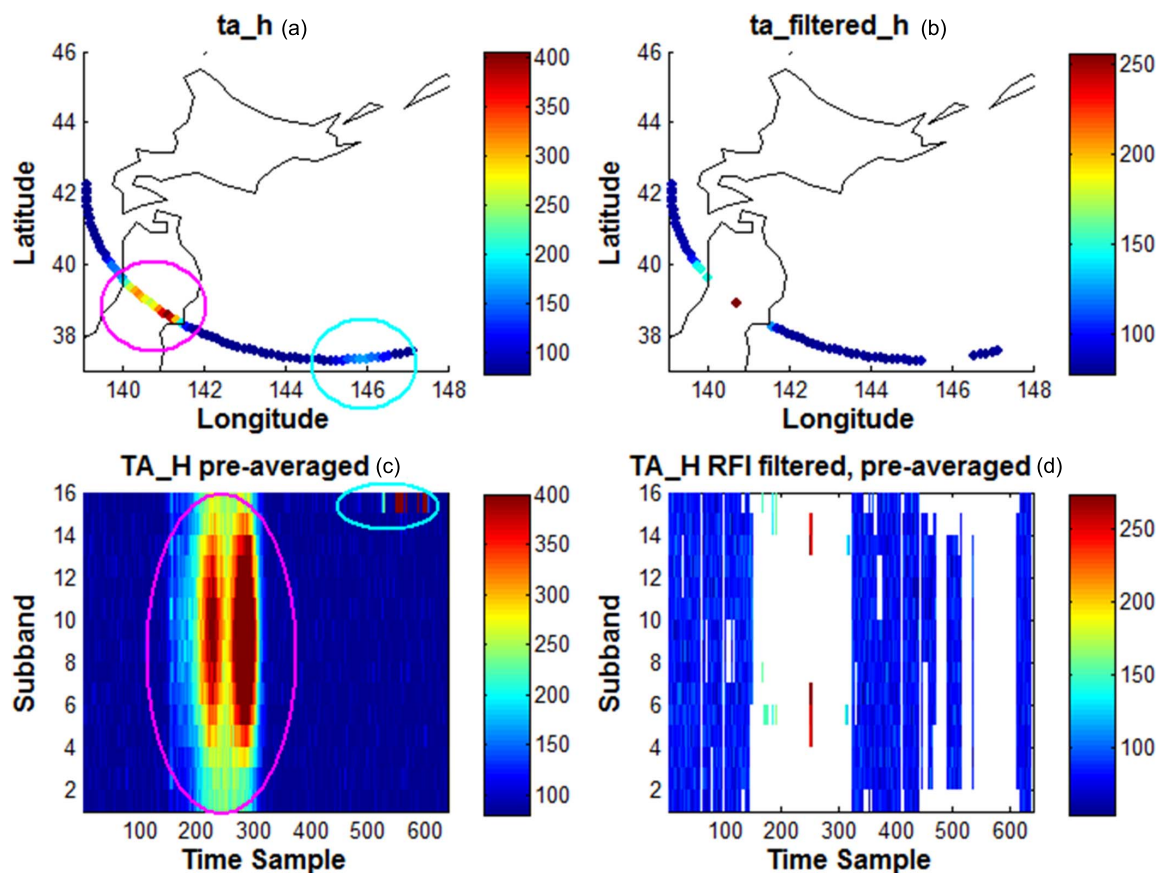


Fig. 9. Plots of the first 80 footprints of horizontally polarized antenna temperatures in kelvins within a scan over Honshu, Japan, taken on June 3, 2015. The figure contains (a) geolocated footprints before RFI mitigation, (b) geolocated footprints after RFI mitigation, (c) corresponding frequency–time data before RFI filtering and footprint averaging, and (d) frequency–time data after RFI filtering.

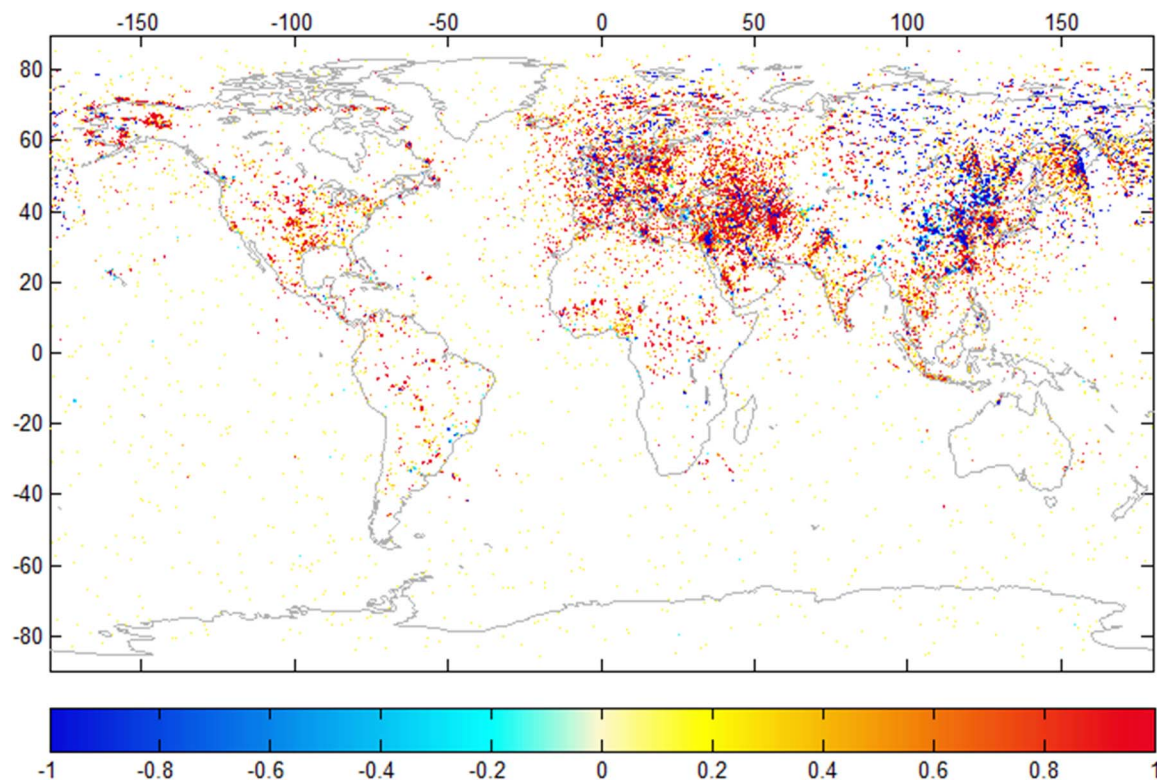


Fig. 10. Fullband excess kurtosis map (kurtosis - 3) of RFI with short (red) and long (blue) duty cycles for the period June 3, 2015 to June 9, 2015.

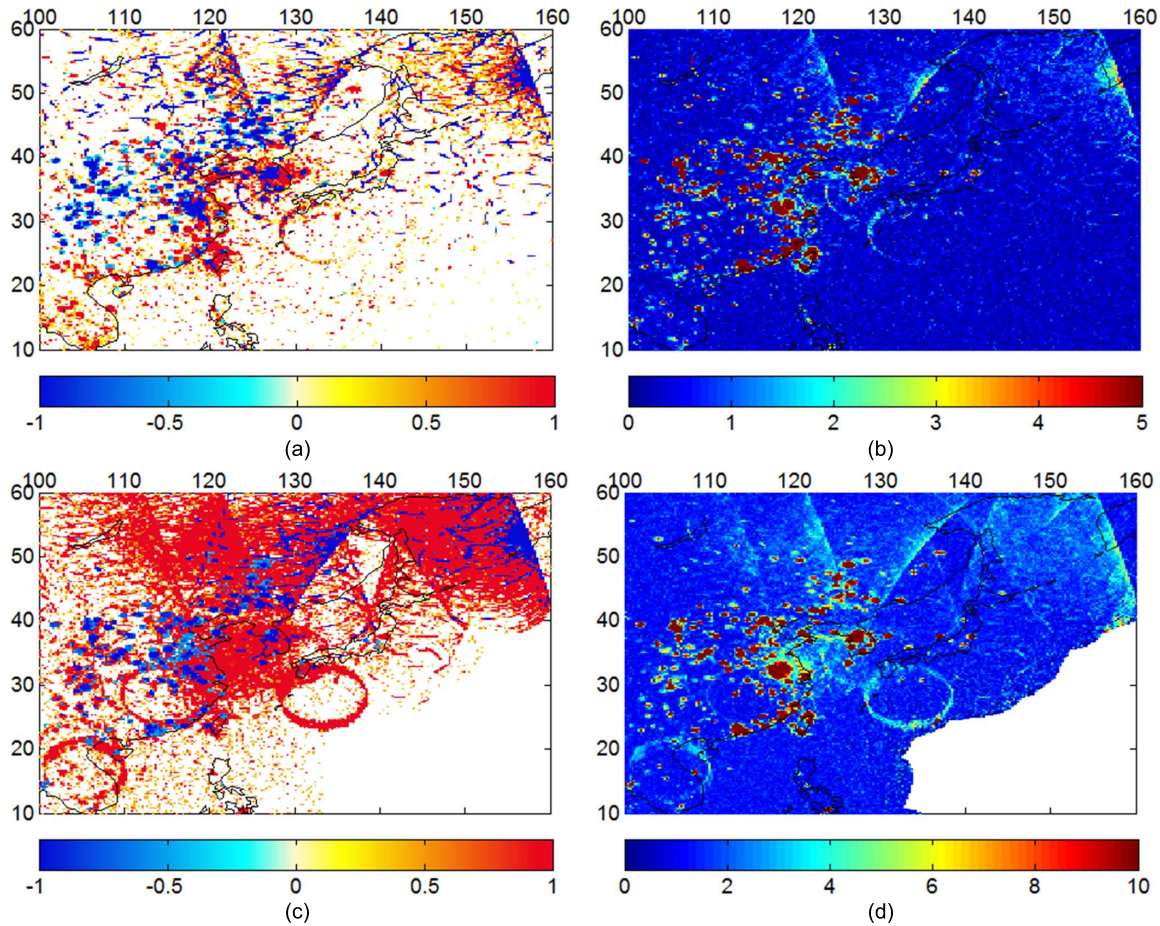


Fig. 11. Excess kurtosis values and corresponding detection rates over East Asia (zoomed-in image of Fig. 10) for the period June 3, 2015 to June 9, 2015. The plot contains the (a) fullband excess kurtosis, the (b) fullband kurtosis detection rate in percent, the (c) subband excess kurtosis, and the (d) subband kurtosis detection rate in percent. The color scales were limited to better show RFI detection.

B. Summary Statistics

Fig. 12 provides additional information on overall brightness temperature and RFI statistics for land-only data from April 2015 to March 2016. Fig. 12(a) shows a complementary cumulative distribution function (CCDF) of horizontally polarized antenna temperatures over one year, including antenna temperatures prior to filtering, after filtering, and after filtering excluding flagged data. The results show the dramatic reduction in antenna temperatures greater than 300 K, indicating successful detection and filtering of large RFI sources. Fig. 12(b) shows a CCDF of the detected RFI level in horizontal polarization. The results show, for example, that about 0.5% of SMAP footprints have RFI levels of 10 K or more, whereas 7% of SMAP footprints have a detected RFI level of 1 K or more. Fig. 12(c) shows a CCDF of the radiometer sensitivity in horizontal polarization (NEDT) following RFI filtering and for footprints without RFI ($t_a - t_{a_filtered} = 0$). The difference between the two curves shows the substantial change in the distribution of NEDT when RFI mitigation occurs. The mean NEDT for all land-only footprints is 1.03 whereas the mean NEDT for land-only footprints with zero RFI level is 0.98. Both meet the radiometric sensitivity requirement of 1.3 K. About 99% of all SMAP land footprints have an NEDT less than 1.5 K,

whereas 0.4% have an NEDT greater than 2 K. Footprints having NEDT greater than or equal to 2 K are flagged and excluded from downstream processing. Finally, Fig. 12(d) shows a CCDF of the percentage of the spectrogram flagged and shows that, for example, approximately 9% of SMAP land-only footprints have 20% or more of the spectrogram flagged. Note that the discretized appearance of this figure for low blanking percentages is due to the finite number (128) of spectrogram points in a single measurement.

Table I specifies the false alarm rates over land for each detector when RFI sources appear to be absent. The false alarm rate (FAR) for the MPD algorithm of 5.66% implies that seven spectrogram pixels are flagged on average in the absence of RFI, resulting in degradation in radiometer sensitivity of the square root of 128/121 or 1.03%. The third Stokes detector FAR is zero due to high thresholds. The pulse blanking, kurtosis, and cross-frequency false alarm rates are near prelaunch expectations with minor differences in the MPD due to altered thresholds increasing the sensitivity of the fourth Stokes detector.

Fig. 13 provides an example illustration of SMAP-observed RFI properties. The color scale of the global map indicates the percentage of the time from April 2015 to March 2016 that the

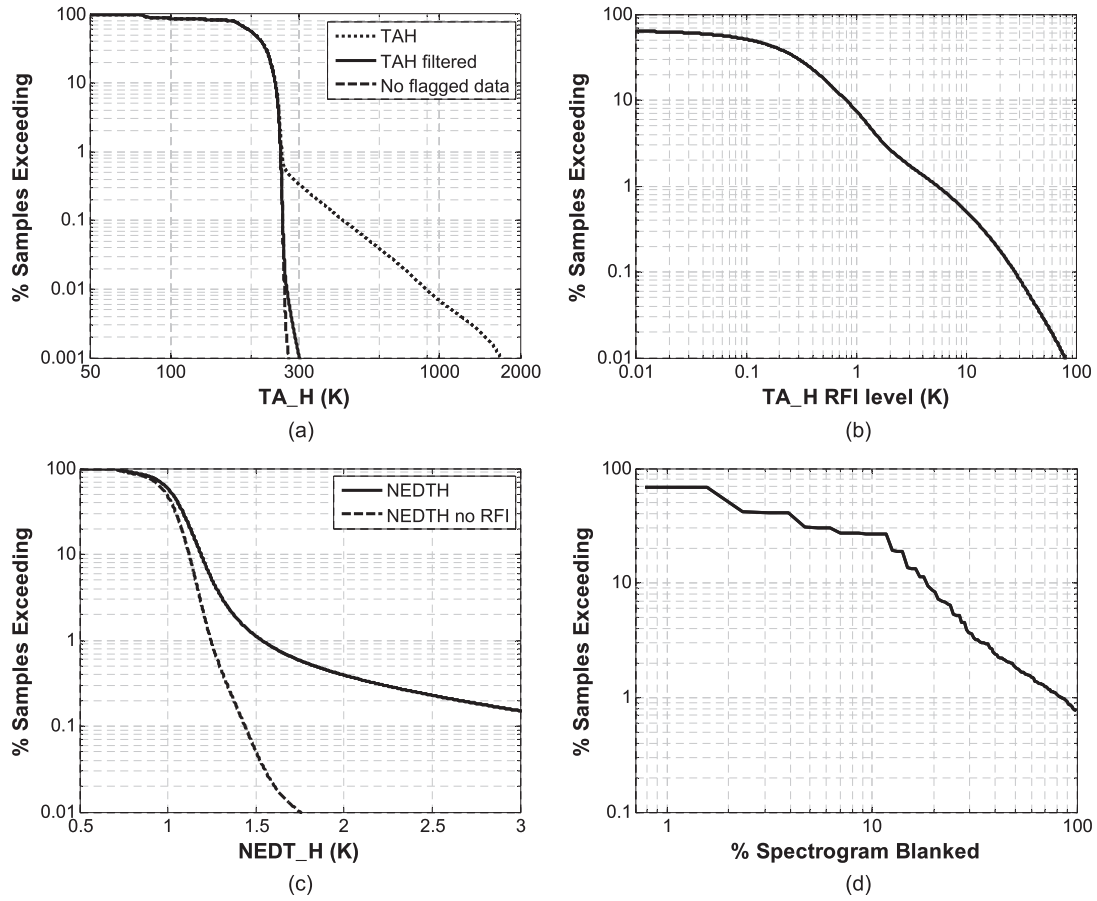


Fig. 12. Summary statistics of SMAP horizontally polarized antenna temperatures for April 2015 to March 2016. (a) CCDF for antenna temperatures before RFI filtering, after RFI filtering, and after RFI filtering and flagging. (b) CCDF of detected RFI levels. (c) CCDF of NEDT following RFI filtering and NEDT of footprints without RFI. (d) CCDF of percentage of spectrogram blanked.

TABLE I
FALSE ALARM RATE (FAR) SUMMARY FOR HORIZONTAL POLARIZATION

Detector	FAR (%)
Pulse Blanking	1.42
Fullband Kurtosis	1.13
Fullband 3 rd Stokes	0
Fullband 4 th Stokes	0.52
1.2 ms Cross Frequency	0.79
9.6 ms Cross Frequency	0.77
Subband Kurtosis	1.11
Subband 3 rd Stokes	0
Subband 4 th Stokes	0.09
MPD	5.66

absolute detected RFI level in horizontal polarization exceeds 5 K. The color scale was limited to 20% to better show variability. When 100% of the time–frequency data are flagged

and blanked due to RFI, the resulting value for the filtered data is a null value indicating a nonvalid result for the observed footprint. RFI in these footprints is also considered to exceed 5 K. Although RFI will always produce a positive bias in the antenna temperatures, the double-sided nature of the time-domain and cross-frequency detectors can produce a negative value in RFI level calculated (the difference between antenna temperatures before and after RFI filtering). There are cases where RFI can bias the robust mean estimates in the detectors, resulting in antenna temperatures which are higher after RFI filtering by greater than 5 K. These footprints are also included in the probability map. The results in Fig. 13 are consistent with previous L-band observations from the SMOS and Aquarius missions, in that RFI is more widespread in Europe and East Asia, particularly in Japan, whereas RFI in other regions appears concentrated in more “point source” locations.

V. CONCLUSION

In general, the results obtained to date indicate that SMAP’s digital back end and associated RFI detection and filtering algorithms are working successfully to improve the quality of SMAP brightness temperature measurements. Results from the comprehensive RFI test conducted at the NASA Goddard Space Flight Center showed empirical and theoretical analyses of the

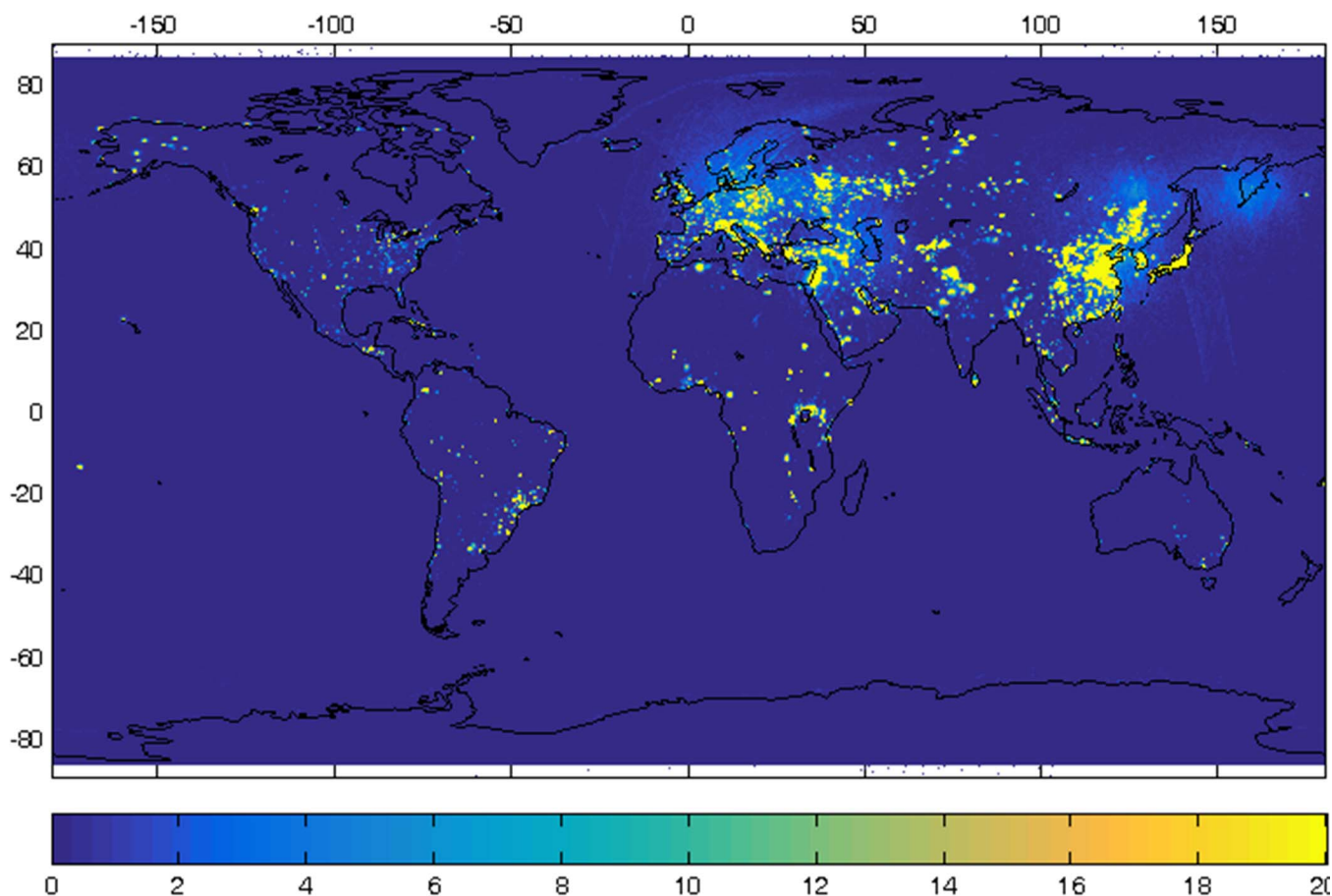


Fig. 13. Percent of the time that SMAP detects an RFI level of 5 K or more in horizontal polarization for data from April 2015 to March 2016.

performance of the SMAP RFI detection algorithms. The ROC and AUC plots, which were used to evaluate detector performance, demonstrate that the cross-frequency algorithms were effective for all the RFI cases considered, whereas the pulse detector and kurtosis algorithms lost their efficiency against continuous and high-duty-cycle pulsed RFI cases. These results are in agreement with that from on-orbit data. The excess kurtosis map shows kurtosis sensitivity to narrow-band sources and sources with low duty cycles such as the radars from the distant early warning line and its insensitivity to wideband RFI sources, which shroud most of Japan.

At launch, detector thresholds were all uniform spatially and temporally. Following preliminary RFI analyses, refinements to the thresholds were made to that of the pulse and fourth Stokes detectors. Both of these algorithms exhibit increased detection rates along coastal crossings. The rapid temporal change in brightness temperatures along a coastal crossing affects the pulse detector, whereas the mismatches in the horizontally and vertically polarized antenna patterns influence the fourth Stokes detector. To avoid excessive false alarms at coastal crossings, the thresholds for these detectors are increased on coastal boundaries. The thresholds for the third Stokes detector remain high, essentially not contributing to the overall false alarm rate. A modification to the third Stokes algorithm to include a specified bias (to account for Faraday rotation) and threshold in the detection algorithm is under consideration. However,

the current algorithms for RFI detection and mitigation are performing as expected. On-orbit results indicate that sources which are wideband and occupy much of SMAP's bandwidth at all times cannot be corrected by any of SMAP's RFI filtering procedures, resulting in the exclusion of extended spatial regions from soil moisture retrievals. These results highlight the need for continued vigilance in the protection of spectrum for microwave radiometry and the need for active enforcement of these protections.

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